MILITARY SPECIFICATION

FOAM, COMBUSTION RETARDANT, FOR CUSHIONING SUPPLY ITEMS

ABOARD NAVY SHIPS

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers inherently combustion retardant flexible foams with low emission of smoke and toxic products to be used for the protection of fragile items aboard Navy ships. These foams include, but are not necessarily limited to polyimide types.

1.2 Classification. The flexible foam should be furnished in the following classes and grades:

Class 1 - Sheet stock, specified thickness
Class 2 - Die cut to specified shape and thickness

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636	- Boxes, shipping, fiberboard

MILITARY

MIL-P-26514	- Polyurethane Foam, Rigid or Flexible for Packaging

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Fitting Out Supply Assistance Team, Naval, Base, Norfolk, VA 23512 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.

DELIVERABLE DATA IS NOT REQUIRED BY THIS DOCUMENT FSC 9330

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STANDARDS

FEDERAL

FED-STD-313 - Material of Safety Data Sheets Preparation and the Submission of

MILITARY

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-759 Tests of Plastics at Subnormal and Supernormal Temperatures.
ASTM D-1596 Shock Absorbing Characteristics of Package Cushioning Materials.
ASTM D-2843 Density of Smoke from Burning or Decomposition of Plastics.
ASTM D-2863 Minimum Oxygen Concentration to Support Combustion.
ASTM D-3574 Flexible Cellular Material Testing.
ASTM I-162 Surface Flammability of Materials.
ASTM F-501 Response to Flame With Vertical Test Specimen.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.2.1 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Dynamic Properties.

The foam shall meet the dynamic cushioning requirements for Type II, Class 2 material as specified in MIL-P-26514, (see 4.3.3.1).

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3.2 Safety for Shipboard Use.

3.2.1 All products furnished under this specification shall be of materials subject to approval by the Navy Environmental Health Center, Norfolk, Virginia. Prospective suppliers shall submit completed Material Safety Data Sheets (OSHA Form 20), (see FED-STD-313 for instructions) and in addition all data and information noted in Table I and herein with their bids.

3.2.1.1 Data and Information to be Submitted.

a. A complete listing of all ingredients contained in the foam is required. All ingredients present in quantities greater than 5% (by weight) must be stated within plus or minus 5%. Those constituents present in quantities less than 5% (by weight) may be estimated to the nearest plus or minus 0.5%, and those materials present in quantities of less than 0.5% (by weight) may be stated as such. Statements such as "does not form hazardous mixtures with other mixtures, solids or gasses" are not acceptable. A full delineation of the foam constituents must be provided.

b. A listing of toxic or potentially toxic gasses/pyrolysis products evolved when the foam is subjected to chemical or thermal degradation is required. Reports of any studies conducted to quantify the nature and magnitude of pyrolysis products evolved during burning of polyimide foam shall also be provided for review. Gaseous products sampled shall include, but not be limited to HF, SO_2, H_2S, CO and HCN.

c. Copies of any available test or other reports relating to adverse health effects which may be experienced by persons engaged in firefighting or damage control operations involving exposure to pyrolysis products from the foam shall be provided.

d. A complete listing (with percentage amounts) of any flame-retardant chemicals utilized in the chemical components together with the percentage of each present in the foam is also required. Of particular concern is the possibility of neurotoxic agents being released from the foam which may constitute a health hazard to personnel from normal foam decomposition products (e.g., carbon monoxide, carbon dioxide, water, hydrogen cyanide, gaseous hydrocarbons, etc.).

3.2.2 Preproduction Testing Requirements. After approval of the Material Safety Data Sheet (OSHA Form 20) and prior to beginning production the contractor shall manufacture preproduction sample quantities of the foam (using the same process by which he will produce the foam to be delivered to the Navy) and perform the test and evaluation described following:

3.2.2.1 Density - The density shall be between 0.4 to 0.8 lbs. per cu. ft. when tested as noted in paragraph 4.3.2.1

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3.2.2.2 Service Temperature - The foam shall remain flexible at all temperatures from - 110° F to 527° F when tested as specified in paragraph 4.3.2.2.

3.2.2.3 Vertical Burn - The flame time shall be 0 seconds, burn length (clear) 1.2 inches maximum and no dripping when a flame is applied as specified in paragraph 4.3.2.3.

3.2.2.4 Smoke Generation - There shall be no more than 4% smoke generation when tested in accordance with paragraph 4.3.2.4.

3.2.2.5 Oxygen Index - The oxygen index shall be no less than 40% when tested as specified in paragraph 4.3.2.5.

3.2.2.6 Flame Spread - The surface flame spread index shall not exceed 1.0 when tested as noted in paragraph 4.3.2.6.

3.2.2.7 Cell Size - Maximum cell diameter shall be 0.20 inch. No more than 25% of the projected area of any cut section of foam shall consist of cells between 0.125 inch diameter and 0.20 inch diameter. All other cells shall be smaller than .0125 when evaluated as described in paragraph 4.3.2.7.

3.2.2.8 Health Hazard. The hazardous products of combustion given off at elevated temperatures shall not exceed the Threshold Limit Values (TLV) for the respective materials as listed by OSHA. When the material is used during normal deployment periods aboard combatant ships, the material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the departmental medical service who will act as an advisor to the contracting agency.

3.3 A detailed report of this testing shall be submitted to the Government for approval. The Government will have 14 days in which to evaluate the data submitted and advise the Contractor of its approval or disapproval.

3.4 Any work performed or material required for the designated production quantities in advance of approval by the Government of testing results shall be at the Contractor's risk.

3.5 Color

All sheets of the foam shall be distinctively marked to distinguish it from other more combustible foams. Marking shall be repeated at 6" spacing and there shall be uniformity of marking from sheet to sheet.

3.6 Sheet Size and Thickness

3.6.1 Class I

Sheet Size	Thickness
4' x 8'	1/2" Thick, Min.
2' x 4'	to
2' x 2'	6" Thick, Max.
	(All Sheet Sizes)

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3.6.2 Class II - Foams shall be sized as specified by the procuring agency. Sizes up to 4' cubes with specified recesses shall be produced by molding or by die cutting and assembly using flexible adhesive.

TABLE I PHYSICAL REQUIREMENTS

PROPERTY	REQUIREMENTS	UNITS	# OF SAMPLES	ASTM TEST	REMARKS
Density	0.4 to 0.8	lbs./ cu. ft.	5	D-3574	Test A only
Service temps.	-110° to +527°	°F	5	D759	Perform D-1596 Per MIL-P-26514
Cushioning properties	must comply w/ MIL-P-26514 at extreme temps.	accelera-tion	5	D-1596	Per Spec. MIL-P-26514 Type II, Class 2
Vertical Burn					
Flame time	0	seconds	3	F-501	
Burn length	1.2 max.	inches	3	F-501	
Dripping	none	-------	3	F-501	
Smoke generation	3% max.	%	7	D-2843	Light absorption
Oxygen index	40% min.	O_2%	7	D-2863	
Flame spread	1.0 max.	index	4	E-162	
Cell Diam.					
Max. Cell Di.	0.20" max.	inches	5	-----	Visual exam. of 1" x 4" Projected area (ink stamped)
from 0.125 to 0.20" Dia.	25% Max.	%	5		

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless noted otherwise in the contract, the contractor is responsible for the performance of all inspection and tests requirements as specified herein. The contractor may use his own or any other facility suitable for the performance of the inspection requirements noted herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to insure conformance. The contractor shall notify the Government in writing at least 5 days before the performance of tests so that the Government may witness these tests at the

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contractors or any other locations according to their prerogative. The samples tested shall be in addition to those delivered to the Government and shall be typical of the material and manufacturing processes of the delivered material. Test results shall be submitted to the Government prior to shipment. (Refer to Paragraph 3.3.2 above). If any of the test samples fail to meet the requirements noted herein the lot shall be rejected.

4.2 Classification of Inspection. The inspections shall be classified as follows:

a. Pre-production inspection and test.

b. Production lot tests.

4.3 Sampling, Test and Inspection.

4.3.1 Lot. For purposes of sampling a lot shall consist of all foam sliced or cut from one foam bun.

4.3.2 Pre-production Inspection and Test. The pre-production (and production) samples shall be taken from different locations throughout the foam bun. The samples shall be representative of the entire pre-production lot. Each sample for any simgle inspection or test shall be foam selected from separated locations within the foam (bun or lot). The inspection and tests to be performed on the pre-production lot shall be as listed in Table I using the following procedures:

4.3.2.1 Foam Density. The density shall be as shown in Table I and para. 3.2.2.1 when five (5) samples are evaluated in accordance with specification ASTM D-3574, Test A. If any sample falls outside the range given in Table I the lot shall be rejected.

4.3.2.2 Service Temperature. The cushioning properties of the foam shall not vary more than ± 10% at -110° and 527°F compared to the same property at room ambient temperatures when tested as specified in MIL-P-26514, for Type II Class 2 foams.

4.3.2.3 Vertical Burn. The flame time, burn length and amount of dripping shall be as noted in Table I and para. 3.2.2.3 when tested in accordance with specification ASTM F-501.

4.3.2.4 Smoke Generation. When tested as specified in ASTM D-2843, there shall be no more light absorbed than noted in Table I and para. 3.2.2.4.

4.3.2.5 Oxygen Index. The limiting oxygen index shall be not less than that noted in Table I and para. 3.2.2.5 when tested by specification ASTM D-2863.

4.3.2.6 Flame Spread. The flame spread index shall be no more than that indicated in Table I and para. 3.2.2.6 when tested as specified in ASTM E-162.

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4.3.2.7 Cell Size. The maximum cell size and the cells larger than 0.125" shall be no more than that shown in Table I and para. 3.2.2.7. When evaluated by visual inspection (and calculation) using a 10 x magnification industrial microscope. Give samples, randomly selected, with smooth cut flat surfaces at least 1" x 4" in size shall be inked and imprinted on smooth white paper for this inspection. If any sample falls outside the maximum cell size or the projected area range (3.2.2.7) the lot shall be rejected (see 6.1).

4.3.2.8 Health Hazard. The toxic products of combustion of the material shall not exceed OSHA standards under normal shipboard conditions, when ignited by hot cigarettes and welding and cutting torch hot metal splatter.

4.3.3 Production Lot Test. Each production lot shall have samples taken in the manner described in para. 4.3.2. If there has been any change in formulation or production methods a new pre-production inspection and tests shall be conducted (see 4.3.2). If there has been no change of formulation or method then the following production lot tests shall be conducted on each production lot.

4.3.3.1 Dynamic Properties. Five samples randomly selected from each lot shall be tested as specified for Type II, Class 2 material in MIL-P-26514. The samples shall comply with the cushioning characteristics for grade A, B, or C foam as specified in the RFP. The densities noted in Figures 4, 5, and 6 of MIL-P-26514 does not necessarily apply to the same densities of these polyimide type foams.

5. PACKAGING

Packaging shall comply with MIL-STD-1188 and shall be adequate for safe shipment, handling and delivery at the destination. Packaging and packing shall be such that the Class I material lays flat. Class II material shall not be compressed or deformed in packaging and packing. Fiberboard cartons shall comply with specification PPP-B-636.

6. NOTES

6.1 If cell size requirements (see 4.3.2.7) can be met by selectively trimming off the out of tolerance material, this action is acceptable as long as sheet size requirements are met.

6.2 Certain polyimide type foams (among others) may be able to comply with the requirements herein.

Custodians:
Navy - SA

Review activities:
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Using activities:
Navy - EC

Preparing activity:
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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